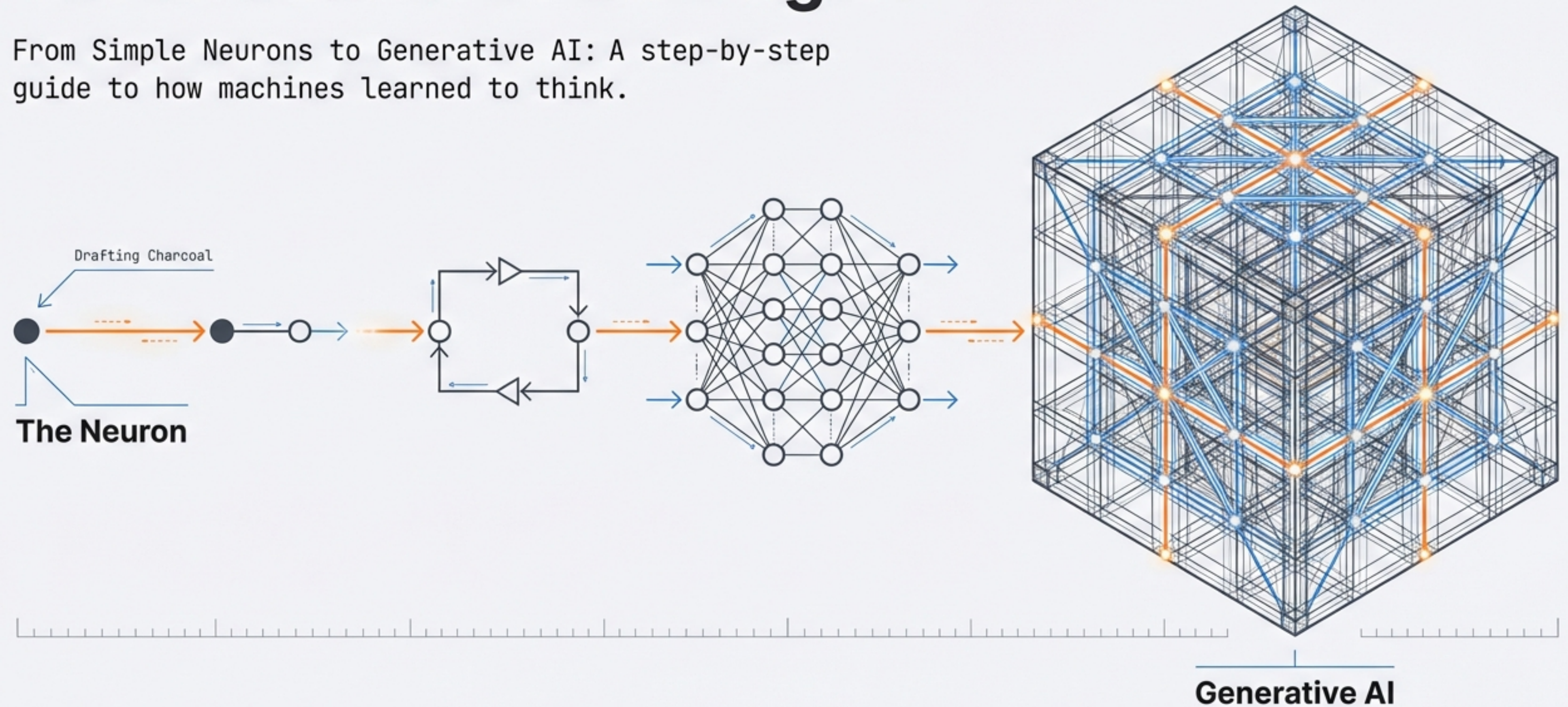
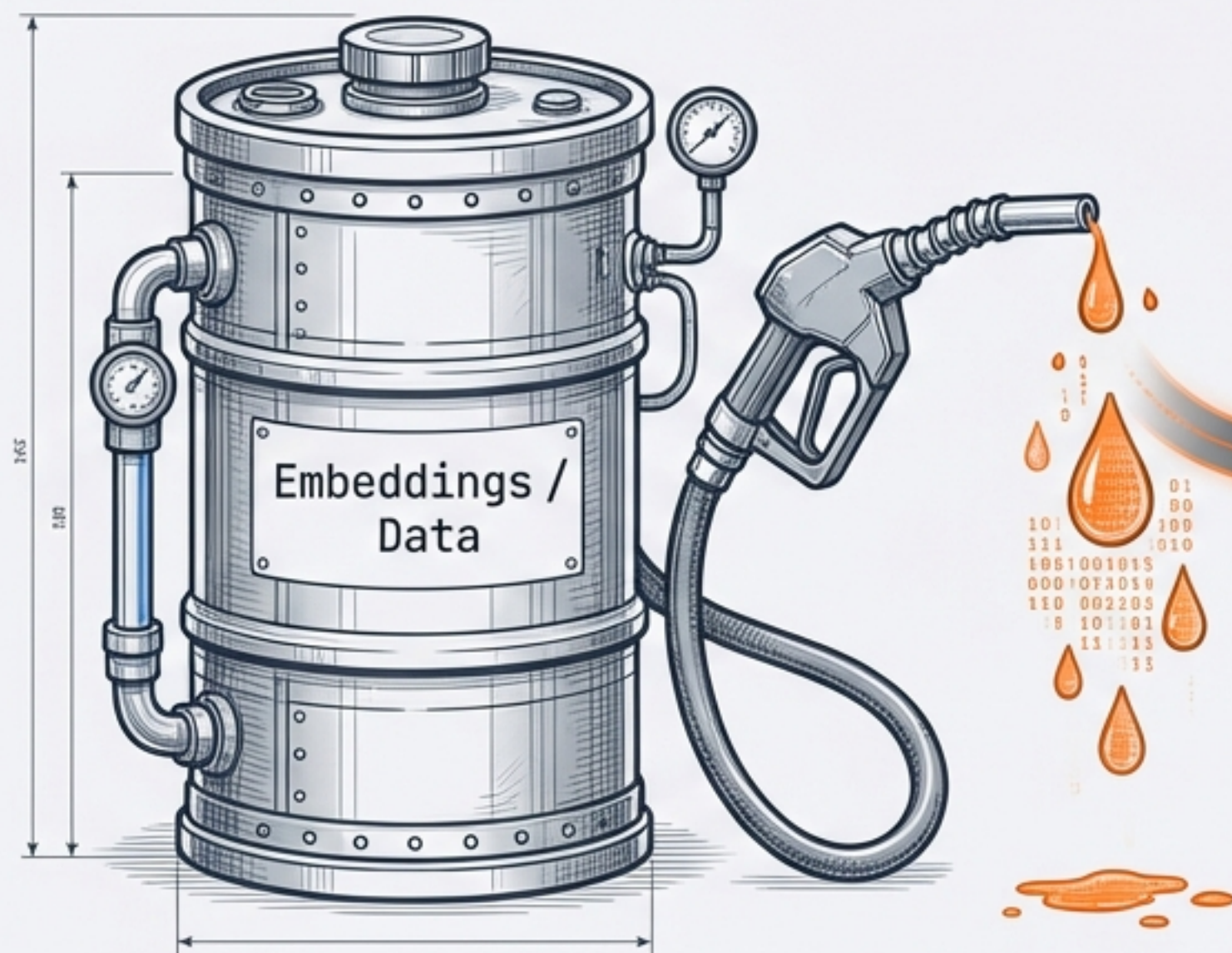


The Evolution of the Engine

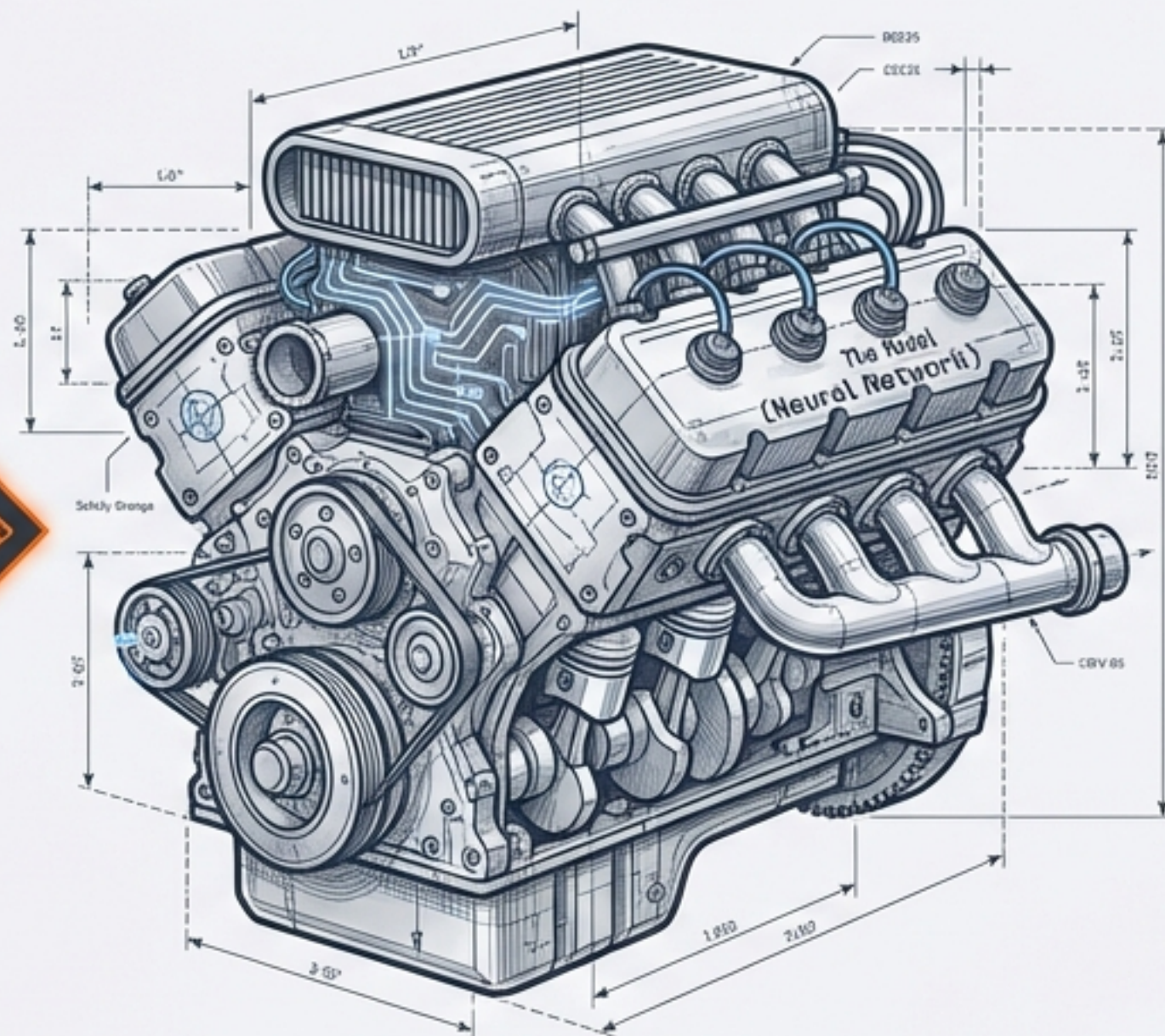
From Simple Neurons to Generative AI: A step-by-step guide to how machines learned to think.



You have the Fuel. Now meet the Engine.

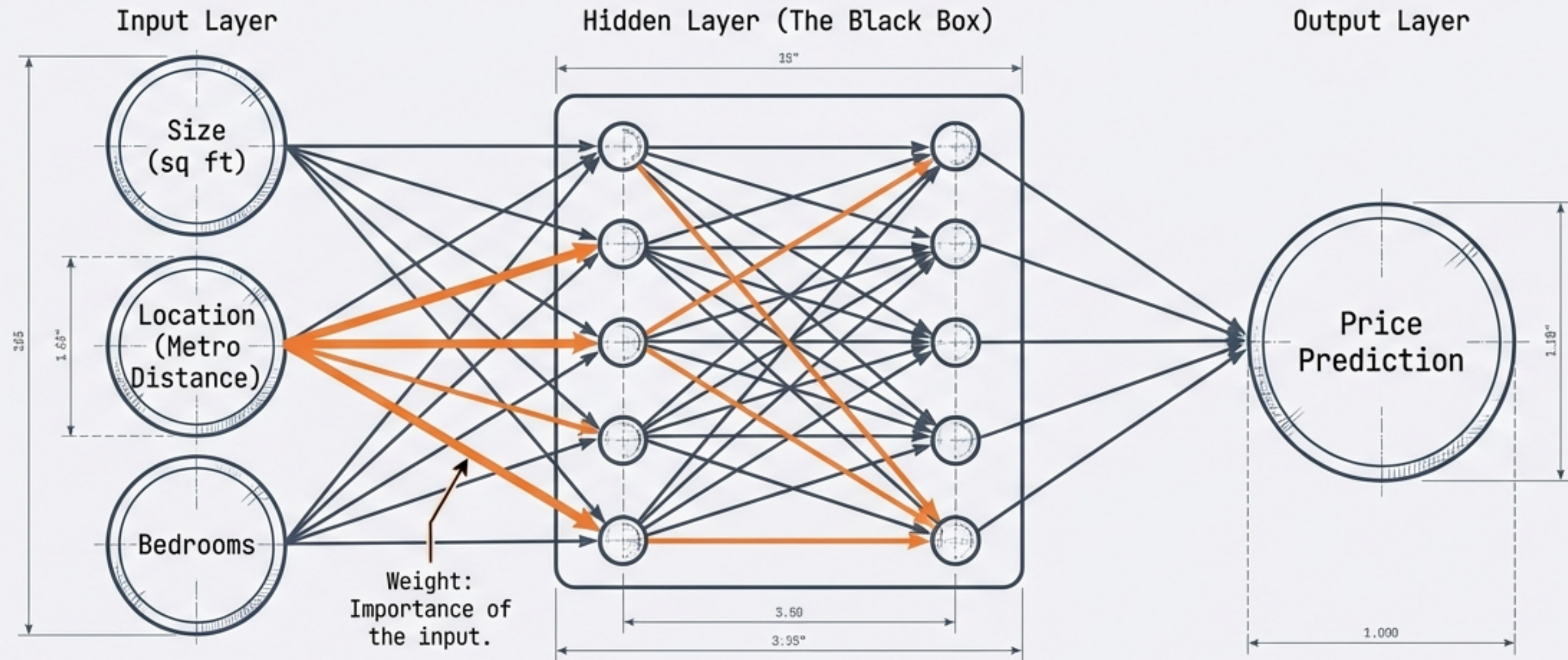


The Fuel: Embeddings turn words into numbers the machine understands.



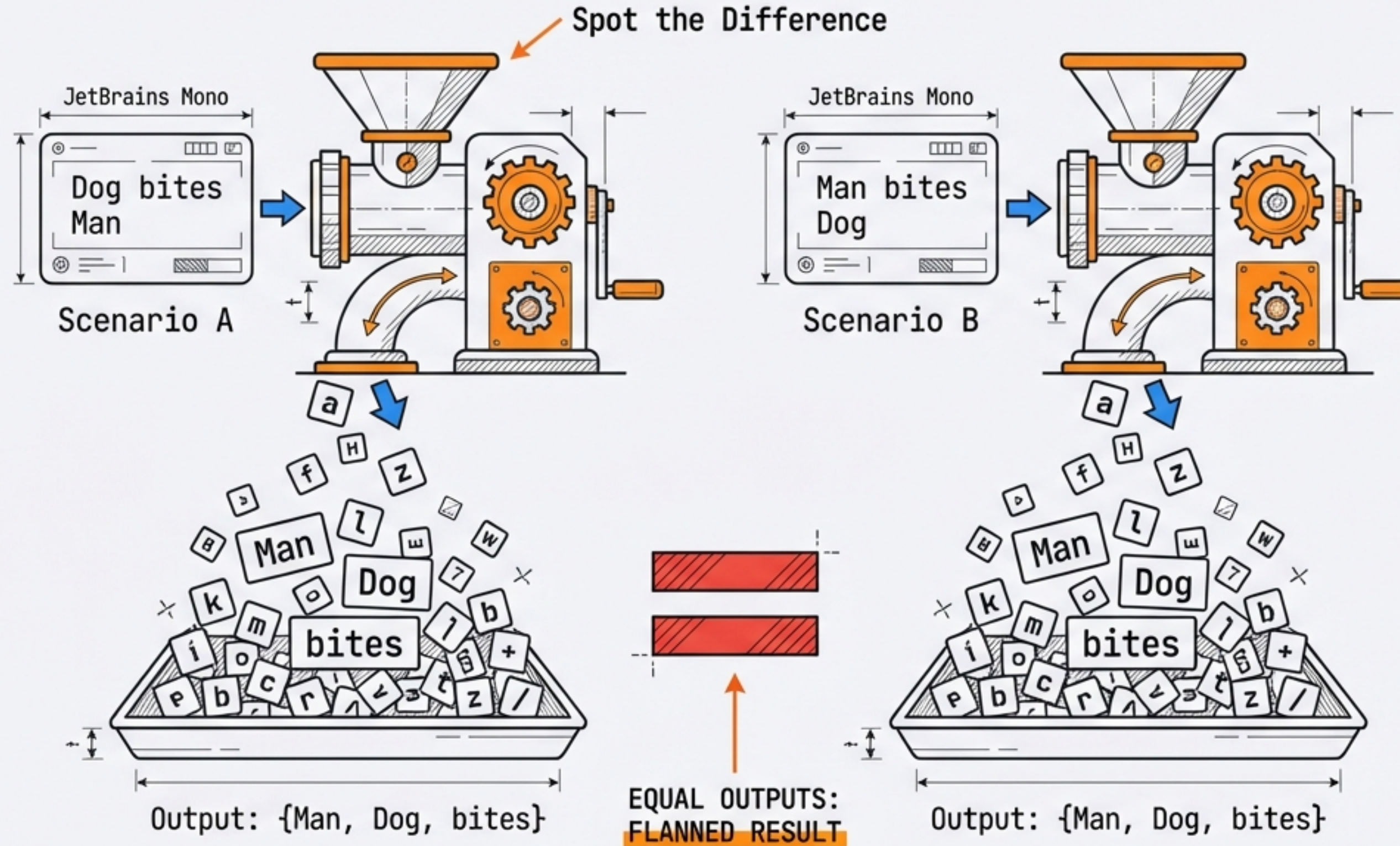
The Engine: The architecture that processes the fuel. To understand ChatGPT, we must first understand the simpler engines that came before it.

Level 1: The Artificial Neural Network (ANN)



ANNs mimic the brain's decision-making process for fixed data tables.

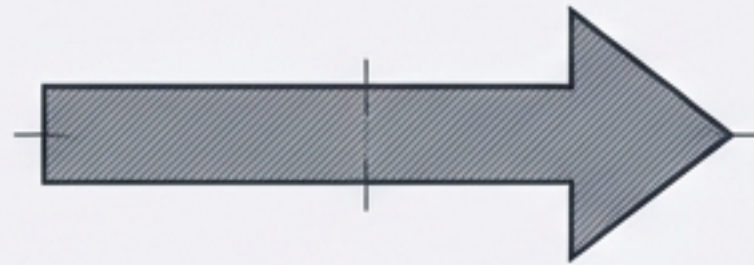
The Flaw: The “Bag of Words” Problem



- Problem 1:**
Fixed Input Size.
ANNs expect rigid
tables, not flexible
sentences.
- Problem 2:**
No Sequence.
The model sees
ingredients but misses
the recipe. It cannot
tell who bit whom.

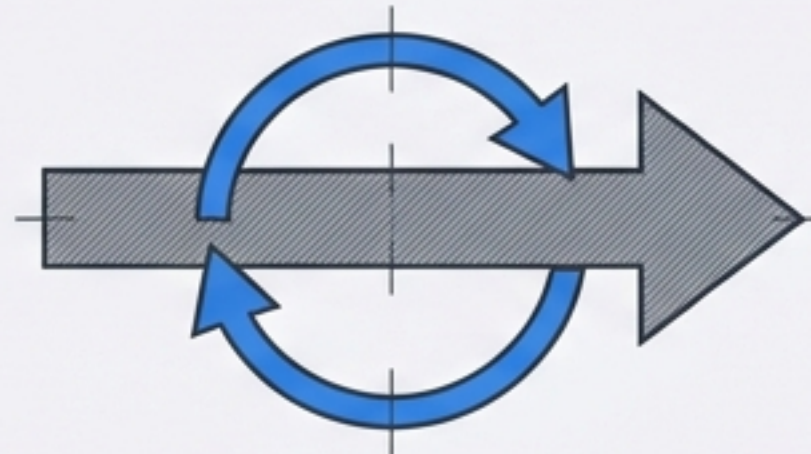
Level 2: The Recurrent Neural Network (RNN)

(ANN)

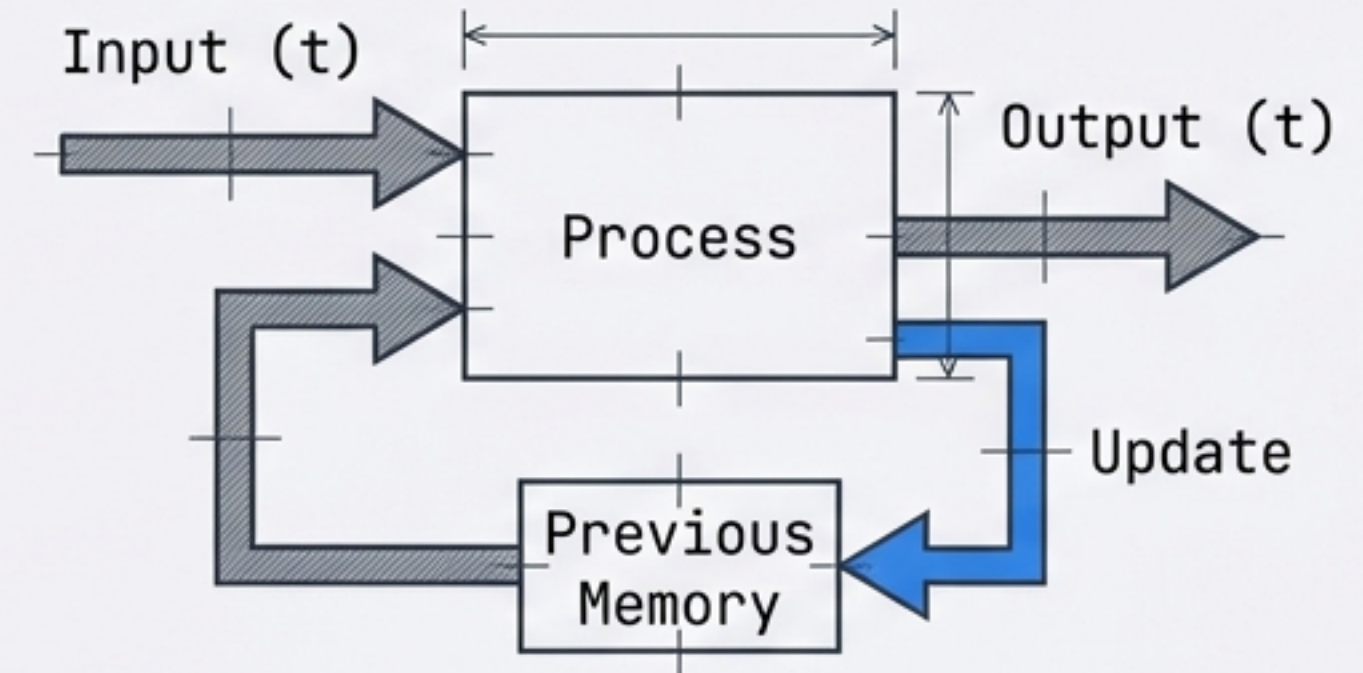


Feed Forward Only

(RNN)



The Feedback Loop



The Fix: Adding Memory. The RNN processes the current word while remembering the "ghost" of the previous word.

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PROJECT:

NEURAL ARCHITECTURE

DWG NO.

AI-ENG-03

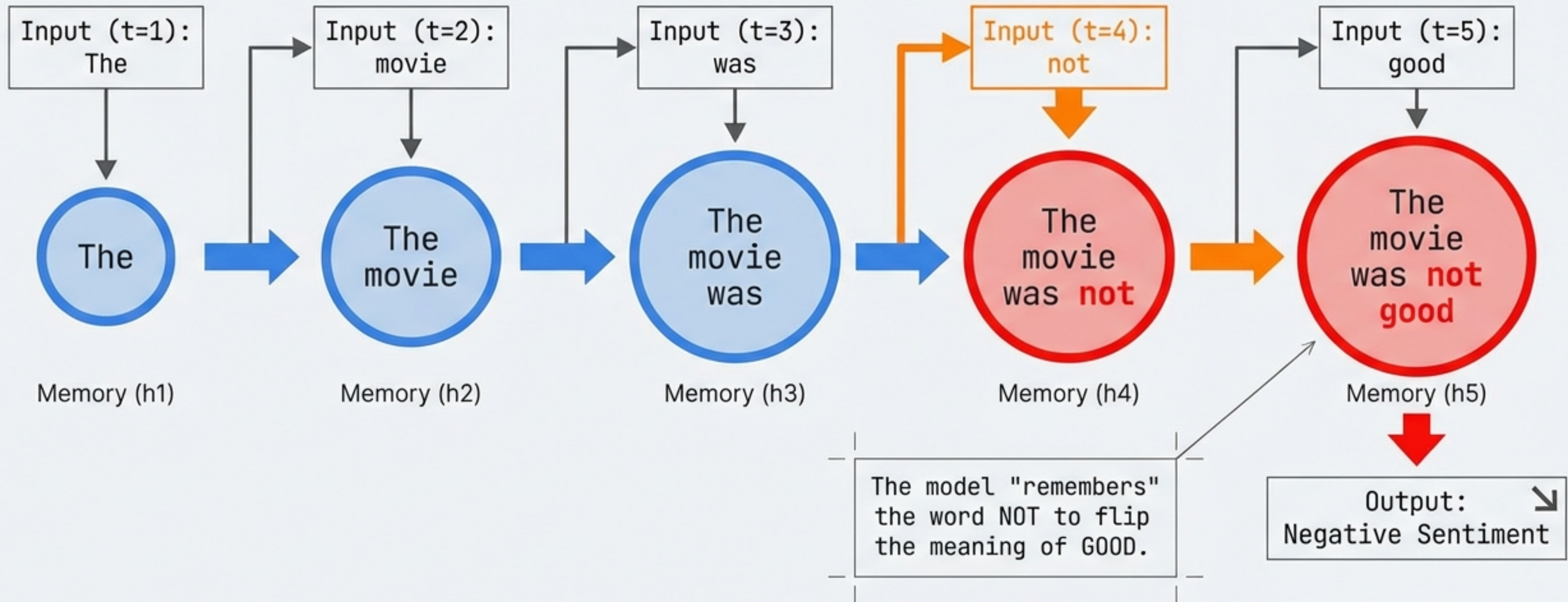
SCALE

1:1

REV

A

RNN in Action: Catching the Negation



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DWG NO. AI-ENG-04

SCALE: 1:1

REV: A

The Flaw: The Goldfish Memory

**Rahul is a
boy...**

...he went to
the market...

...he refers to
himself as...



The Vanishing Gradient Problem.

RNNs have severe short-term memory. By the time it reaches the end of a paragraph, the signal from the beginning has faded away.

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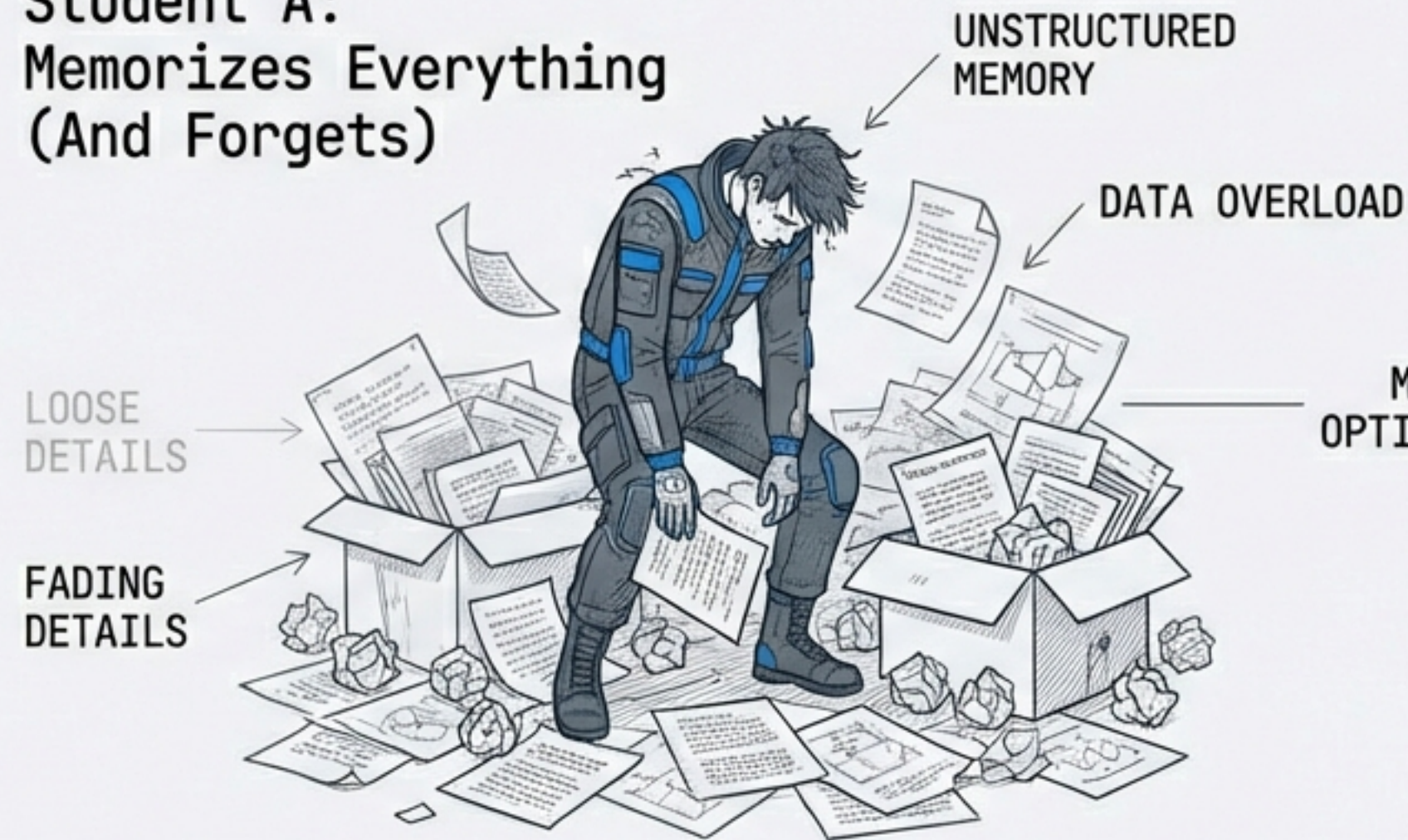
DWG NO. AI-ENG-05

SCALE: 1:1

REV: A

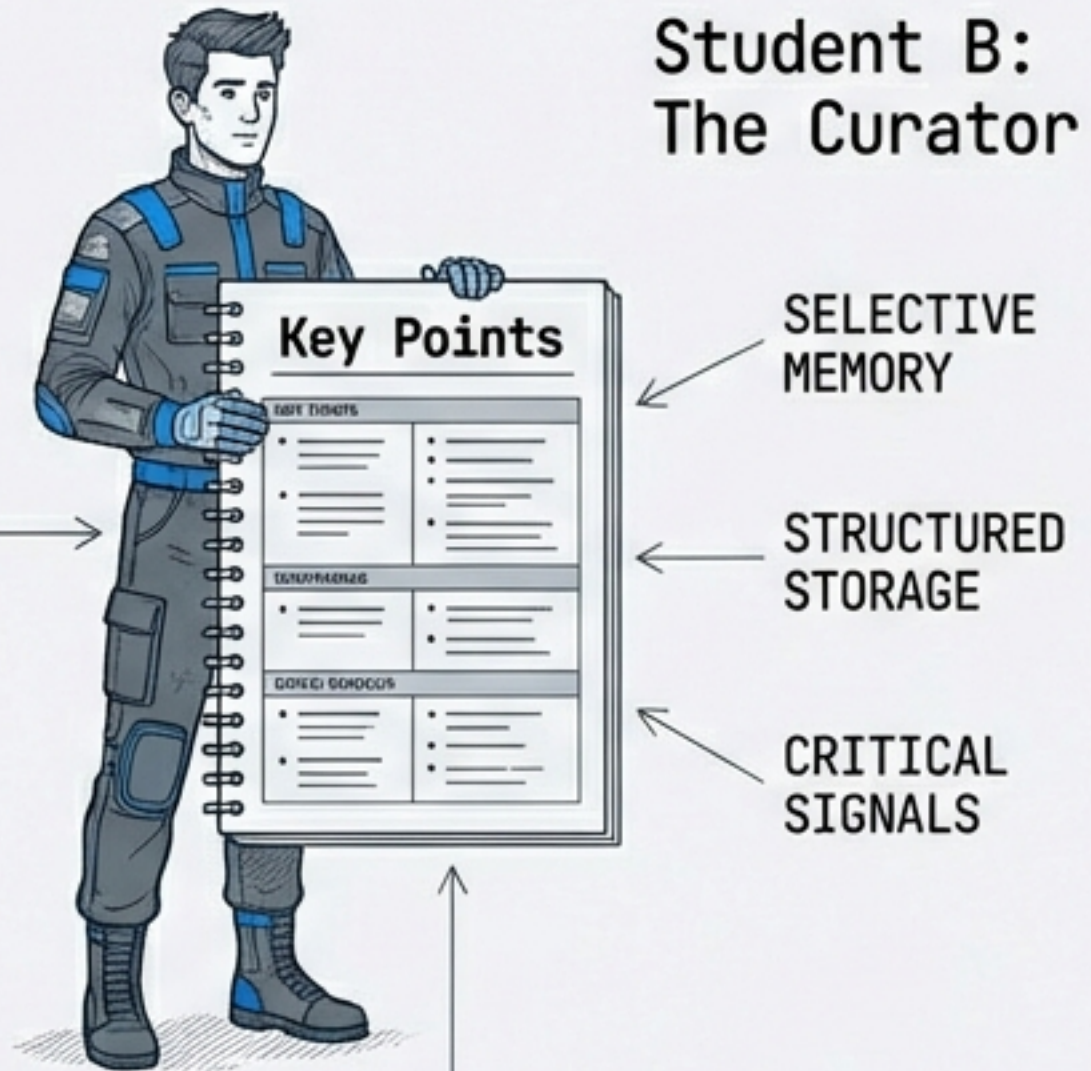
Level 3: Long Short-Term Memory (LSTM)

Student A:
Memorizes Everything
(And Forgets)



MEMORY
OPTIMIZATION

Student B:
The Curator



The Fix: The Notebook.

Instead of trying to remember everything, the LSTM learns what to write down and what to ignore.

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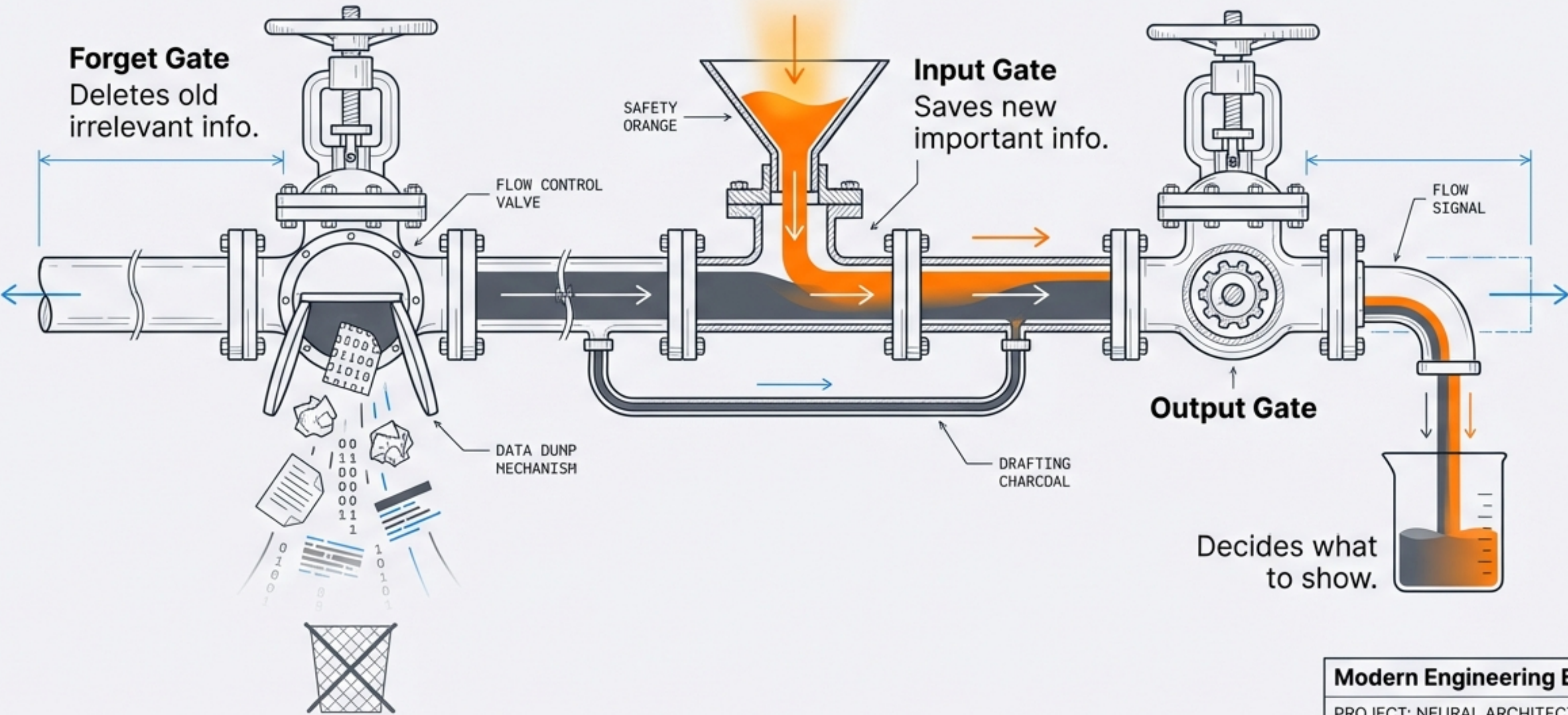
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DWG NO. AI-ENG-06

SCALE: 1:1

REV. A

Inside the LSTM: The Three Gates



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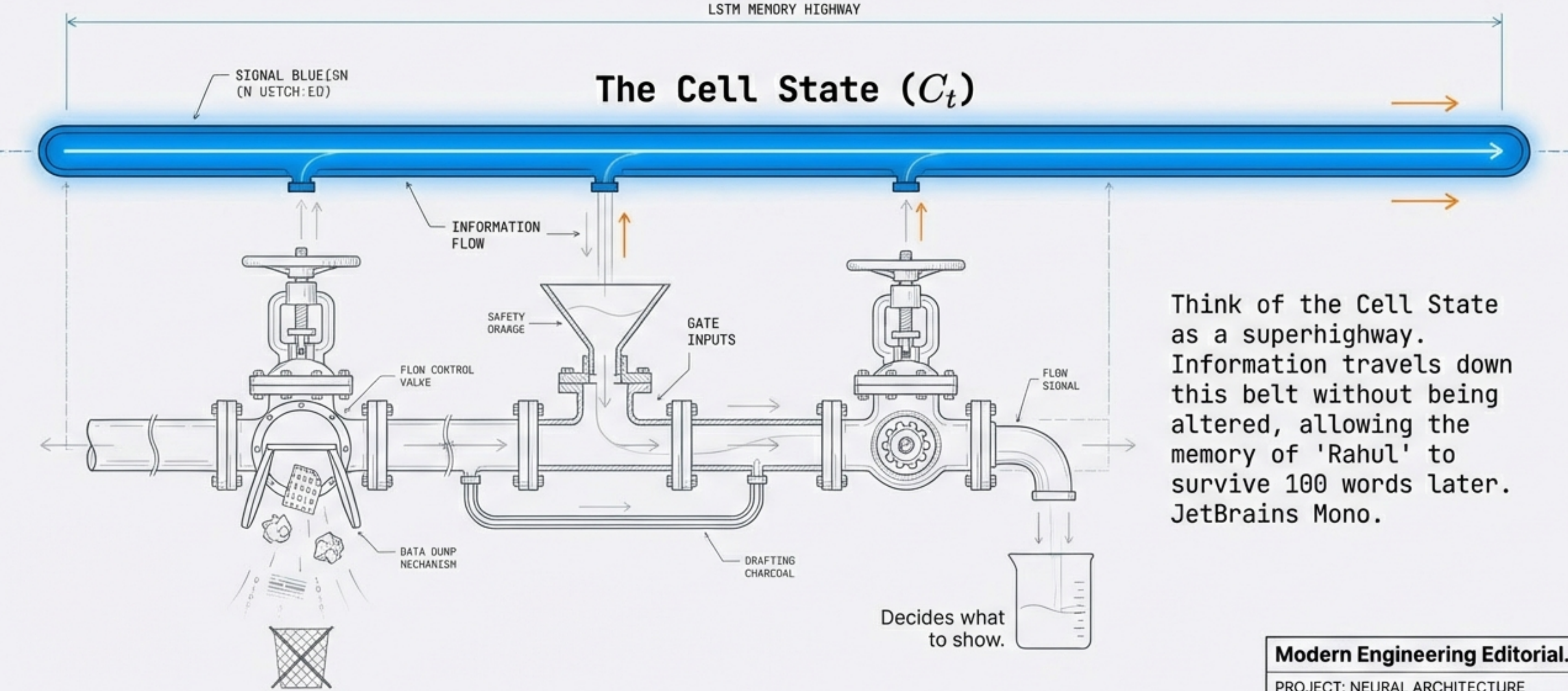
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DWG NO. AI-ENG-07

SCALE: 1:1

REV. A

The Secret Sauce: The Conveyor Belt

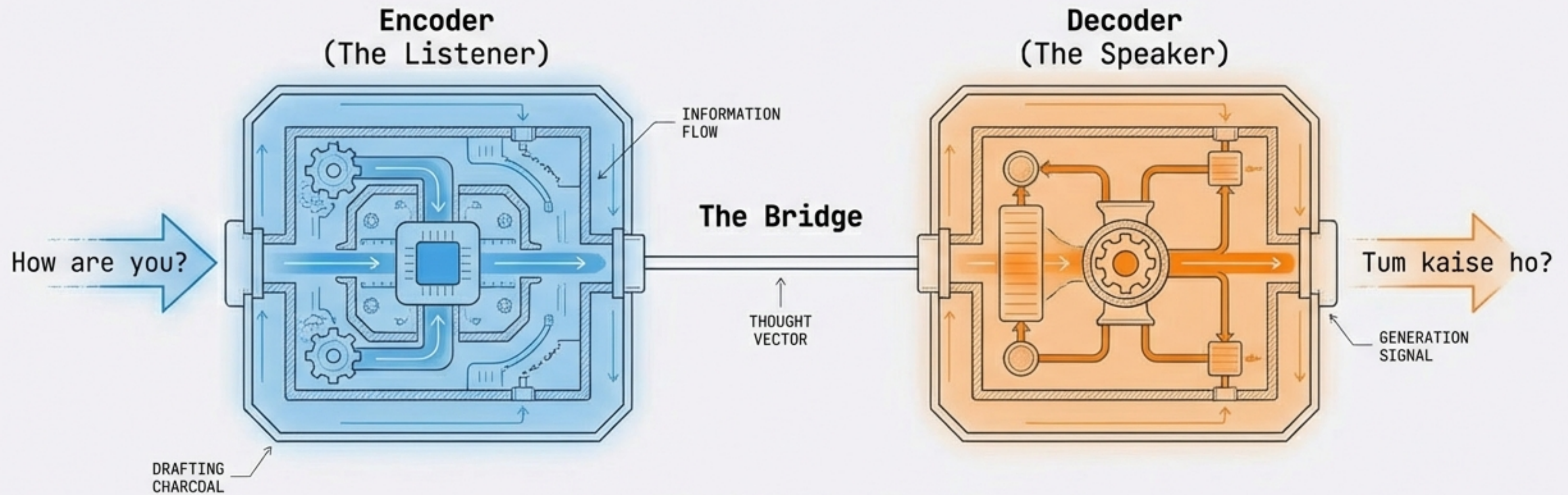


Think of the Cell State as a superhighway. Information travels down this belt without being altered, allowing the memory of 'Rahul' to survive 100 words later. JetBrains Mono.

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PROJECT: NEURAL ARCHITECTURE		
DWG NO. AI-ENG-08	SCALE: 1:1	REV. A

Level 4: The Encoder-Decoder (Seq2Seq)

The Application: Machine Translation.



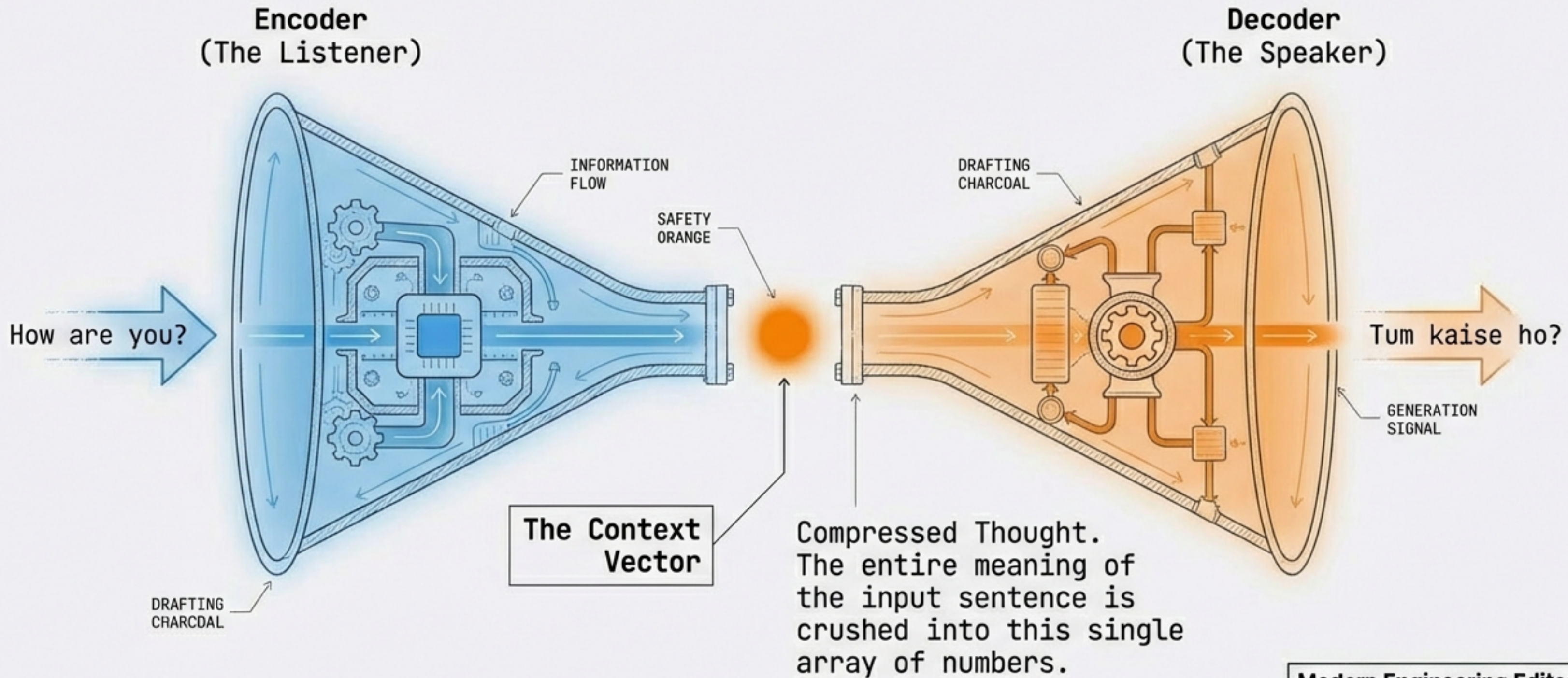
We chain two LSTMs together.
One listens to English,
passes the thought,
and the other speaks Hindi.

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PROJECT: NEURAL ARCHITECTURE

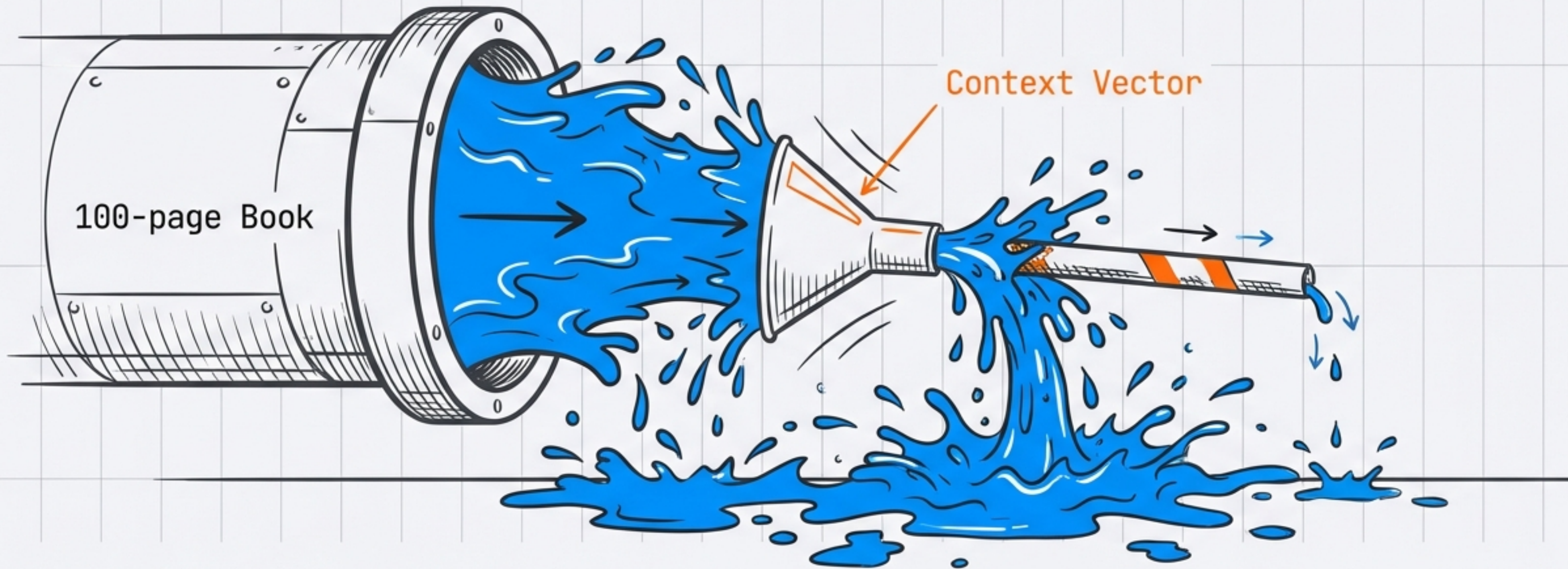
DWG NO. AI-ENG-09 | SCALE: 1:1 | REV. A

The Process: Crushing the Thought



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DWG NO. AI-ENG-10	SCALE: 1:1	REV. A

The Flaw: The Information Bottleneck



Imagine summarizing a whole book into one sentence. Details are lost. The Encoder cannot squeeze all the nuance of a long sequence into a single vector.

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AI-ENG-11

SCALE:
1:1

REV.
A

The Solution: Looking Back

Input Sequence

How
are
you
today?

Attention Mechanism

Output Sequence

Tum
kaise
ho
aaj?

We allow the Decoder to 'look back' at the entire source material. This simple shift birthed the Transformer (ChatGPT).

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NEURAL ARCHITECTURE

DWG NO.
AI-ENG-12

SCALE:
1:2

REV.
A

The Evolutionary Cheatsheet

Era	Model		Strength	The Fatal Flaw
1	ANN		Fixed Numbers	No Sequence (Bag of Words)
2	RNN		Sequences	Goldfish Memory
3	LSTM		Long Memory	Slow (Sequential)
4	Seq2Seq		Translation	Bottleneck
5	Transformers		Everything	None of the above